1 Write down the next two terms in the following quadratic sequence.

9 13 19 27

+4 +6 +8

2 2

27 + 10 = 37

37 + 12 = 49

37 , 49

(Total for Question 1 is 2 marks)

2 Write down the next two terms in the following quadratic sequence.

-5 0 9 22

+5 +9 +13

4 4

22 + 17 = 39

39 + 21 = 60

39 , 60

(Total for Question 2 is 2 marks)

3 The nth term of a sequence is

$$2n^2 + 4n - 1$$

Work out the 10th term of the sequence.

$2(10)^2 + 4(10) - 1$

239

(Total for Question 3 is 2 marks)

4 The nth term of a sequence is

$$n^2 + 2n$$

Work out the first 5 terms of the sequence.

$(1)^2 + 2(1) = 3$

$(2)^2 + 2(2) = 8$

$(3)^2 + 2(3) = 15$

$(4)^2 + 2(4) = 24$

$(5)^2 + 2(5) = 35$

3, 8, 15, 24, 35

(Total for Question 4 is 2 marks)

5 Here are the first 5 terms of a quadratic sequence.

5 11 19 29 41

Find an expression, in terms of n, for the nth term of this sequence.

$an^2 + bn + c$

$a + b + c \rightarrow$ 5 11 19 29 41

$3a + b \rightarrow$ 6 8 10 12

$2a \rightarrow$ 2 2 2

$2a = 2$

$a = 1$

$3a + b = 6$

$3(1) + b = 6$

$3 + b = 6$

$b = 3$

$a + b + c = 5$

$1 + 3 + c = 5$

$4 + c = 5$

$c = 1$

$n^2 + 3n + 1$

(Total for Question 5 is 4 marks)

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6 Here are the first 5 terms of a quadratic sequence.

2 10 22 38 58

Find an expression, in terms of n, for the nth term of this sequence.

2 10 22 38 58

8 12 16 20

4 4 4

$2a = 4$

$a = 2$

$3a + b = 8$

$3(2) + b = 8$

$6 + b = 8$

$b = 2$

$a + b + c = 2$

$2 + 2 + c = 2$

$4 + c = 2$

$c = -2$

$2n^2 + 2n - 2$

(Total for Question 6 is 4 marks)

7 Here are the first 5 terms of a quadratic sequence.

15 19 25 33 43

Find an expression, in terms of n, for the nth term of this sequence.

15 19 25 33 43

4 6 8 10

2 2 2

$2a = 2$

$a = 1$

$3a + b = 4$

$3(1) + b = 4$

$3 + b = 4$

$b = 1$

$a + b + c = 15$

$1 + 1 + c = 15$

$c = 13$

$n^2 + n + 13$

(Total for Question 7 is 4 marks)

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8 Here are the first 5 terms of a quadratic sequence.

2 10 24 44 70

Find an expression, in terms of n, for the nth term of this sequence.

2 10 24 44 70

8 14 20 26

6 6 6

$2a = 6$

$a = 3$

$3a + b = 8$

$3(3) + b = 8$

$9 + b = 8$

$b = -1$

$a + b + c = 2$

$3 - 1 + c = 2$

$2 + c = 2$

$c = 0$

$3n^2 - n$

(Total for Question 8 is 4 marks)

9 Here are the first 5 terms of a quadratic sequence.

19 15 9 1 -9

Find an expression, in terms of *n*, for the *n*th term of this sequence.

19 15 9 1 -9

-4 -6 -8 -10

-2 -2 -2

$2a = -2$

$a = -1$

$3a + b = -4$

$3(-1) + b = -4$

$-3 + b = -4$

$b = -1$

$a + b + c = 19$

$-1 - 1 + c = 19$

$-2 + c = 19$

$c = 21$

$-n^2 - n + 21$

(Total for Question 9 is 4 marks)

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10 Here are the first 5 terms of a quadratic sequence.

-2 -1 1 4 8

Find an expression, in terms of *n*, for the *n*th term of this sequence.

-2 -1 1 4 8

1 2 3 4

1 1 1

$2a = 1$

$a = 0.5$

$3a + b = 1$

$3(0.5) + b = 1$

$1.5 + b = 1$

$b = -0.5$

$a + b + c = -2$

$0.5 - 0.5 + c = -2$

$c = -2$

$0.5n^2 - 0.5n - 2$

(Total for Question 10 is 4 marks)

11 Here are the first 5 terms of a quadratic sequence.

6 10 16 24 34

(a) Show that the nth term is $n^2 + n + 4$

6 10 16 24 34

4 6 8 10

2 2 2

$2a = 2$

$a = 1$

$3a + b = 4$

$3(1) + b = 4$

$3 + b = 4$

$b = 1$

$a + b + c = 6$

$1 + 1 + c = 6$

$2 + c = 6$

$c = 4$

$n^2 + n + 4$ **(4)**

(b) Hence, determine whether 136 is a term in the sequence.

$n^2 + n + 4 = 136$

$n^2 + n - 132 = 0$

$(n + 12)(n - 11) = 0$

$n = -12 \quad n = 11$

136 is the 11th term

(2)

(Total for Question 11 is 6 marks)

12 Here are the first 5 terms of a quadratic sequence.

-8 2 16 34 56

(a) Show that the nth term is $2n^2 + 4n - 14$

-8 2 16 34 56

10 14 18 22

4 4 4

$2a = 4$

$a = 2$

$3a + b = 10$

$3(2) + b = 10$

$6 + b = 10$

$b = 4$

$a + b + c = -8$

$2 + 4 + c = -8$

$6 + c = -8$

$c = -14$

$2n^2 + 4n - 14$

(4)

(b) Hence, determine whether 272 is a term in the sequence.

$2n^2 + 4n - 14 = 272$

$2n^2 + 4n - 286 = 0$

$n^2 + 2n - 143 = 0$

$(n + 13)(n - 11) = 0$

$n = -13 \quad n = 11$

272 is the 11th term

(2)

(Total for Question 12 is 6 marks)